

COURSE DESCRIPTION			
Program	Automation and Robotics		
Course Title	Foundations of Robotics		
Course Code	2331	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course provides a comprehensive introduction to the fundamentals of robotics. It begins by defining what robots are and exploring their historical development and key characteristics. The course then examines different classifications of robots based on their structure, functionality, and application areas, such as industrial, service, and mobile robots.

Course Objectives

1	To explore the broad scope and applications of robotics in various domains.
2	To understand the functional elements and building blocks of robotic systems.
3	To study sensors and actuators used in robotic applications.
4	To develop an understanding of basic autonomous and intelligent robotic systems.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in Physics	2002B	Engineering Physics for Applied Electrical Technology and Computing	I
Basic knowledge in Chemistry	2003A	Chemistry for Engineering Practices	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Select and use sensors in various robotic application contexts	Apply
CO2	Select and use actuators in various robotic application contexts	Apply
CO3	Explain the broad concepts of robotics and its applications	Understand
CO4	Develop basic kind of autonomous robot	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	-	3	2	-	-	-	-	-	-
CO2	2	3	3	2	2	-	-	-	-	-	-
CO3	2	2	-	3	3	3	-	-	-	-	3
CO4	3	3	3	2	3	3	-	-	-	-	3
Total	10	10	6	10	10	6	-	-	-	-	6
Correlation Level	2.5	2.5	3	2.5	2.5	3	-	-	-	-	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Sensors in Robotics	Basics of Sensors, Internal Sensors	15	0
Module II	Actuators, Drives and Advanced Robotics Applications	Building Blocks of a Robot	17	0
Module III	Fundamentals of Robotics & Applications	Definitions of Robots and Robotics	13	0
Module IV	Autonomous Robots - Line Follower and Obstacle Avoider	Basic Principle of Line Follower Robot	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Basics of sensors	Explain the concept and importance of sensors in robotics	L	CO1	Understand	4
Internal sensors: position and velocity sensor	Explain working and selection of position and velocity sensors	L	CO1	Apply	2
External sensors: proximity, force and torque sensors	Explain working and application of external sensors	L	CO1	Apply	2
Sensing variables in robotics	Identify different sensing variables and suitable sensors	L	CO1	Understand	3
Variable recognition systems	Explain basic principles of recognition systems in robotics	L	CO1	Apply	4
Module II					
Building blocks of a robot	Explain the building blocks of a robotic system	L	CO2	Understand	2
Actuators and DC motors	Explain types of DC motors used as actuators	L	CO2	Understand	3
Speed and direction control of drives	Apply speed and direction control techniques for motors	L	CO2	Apply	3
Non-traditional sensors and actuators	Explain non-traditional sensors and actuators	L	CO2	Apply	2
Advanced sensors (optical, inertial, biosensors)	Apply advanced sensors in robotic applications	L	CO2	Apply	3
Selection of sensors and actuators for self-driving car	Select appropriate sensors and actuators for autonomous vehicles	L	CO2	Apply	4
Module III					
Fundamentals of robots: definition, need and types	Explain the basic definition, need, and classification of robots	L	CO3	Understand	1
Characteristics of robots	Describe key characteristics and performance parameters of robots	L	CO3	Understand	2
Applications of robots	Identify applications of robots in medical, space, defense, domestic and entertainment sectors	L	CO3	Understand	3
Industrial applications of robots	Explain the role of robots in material handling, welding, painting and machining	L	CO3	Understand	3
Parts and functions of robots	Describe major parts of a robot and explain their functions	L	CO3	Understand	4
Module IV					
Principle of line follower robot	Explain the working principle of a line follower robot	L	CO4	Apply	2
Assembling sensors	Demonstrate the assembly of sensors used in line follower robots	L	CO4	Apply	3

Assembling line follower robot	Assemble a complete line follower robotic system	L	CO4	Understand	2
Interfacing microcontroller with motor driver and sensors	Explain interfacing of microcontroller with motor driver and sensor IC	L	CO4	Remember	3
Assembling obstacle avoiding robot	Assemble and test an obstacle avoiding robot	L	CO4	Remember	2
Principle of robot motion	Explain the basic principle of robotic movement	L	CO4	Remember	3

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare notes on internal sensors such as position, velocity sensors, and accelerometers with applications.	6
2	Study external sensors including proximity, force, and torque sensors and identify their real-time uses in robotics.	6
3	Create a table mapping different sensing variables (temperature, light, humidity, sound) with suitable sensors.	6
4	Prepare chart for all sensors and take a seminar each sensors	6
Module II		
5	Study different types of electric motors (DC, servo, stepper) used as actuators in robotic systems.	6
6	Apply speed and direction control techniques of motors and selection criteria for actuators in robots	6
7	Case study on selection of sensors and actuators for self-driving cars and maze-solving robots (advanced content).	6
Module III		
8	Study and summarize the definitions, need, types, and characteristics of robots using textbook and NPTEL resources.	6
9	Prepare a comparative chart on advantages and limitations of robots in different application areas such as medical, defense, and domestic sectors.	6
10	Explore real-world industrial robot applications (material handling, welding, painting) through videos and case studies.	6
11	Prepare report on case studies based robot applications	6
Module IV		
12	Study the basic principle of line follower robots and color sensing using online simulations and reference videos.	6
13	Identify different sensors used in line follower and obstacle avoiding robots and explain their working principles.	6
14	Analyze the block diagram of a line follower robot and explain signal flow from sensor to actuator	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Robotics for Engineers	Yoram Koren	McGraw-Hill Education
2	Industrial Robotics: Technology, Programming and Applications	Mikell P. Groover	McGraw-Hill Education
3	Textbook on Industrial Robotics	G. S. Hedge	Laxmi Publications

REFERENCE			
Sl. No.	Title	Author	Publication
1	Mechanical Measurements and Instrumentation	A. K. Sawhney & Puneet Sawhney	Dhanpat Rai & Co.
2	Robotics	K. K. Appu Kuttan	I.K. International

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://nptel.ac.in	NPTEL Robotics Courses
2	https://create.arduino.cc/projecthub	Arduino Project

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60